

Data Science - Research Assignment 1

SECTION A.

- ① A database is an organised collection of data stored electronically so it can be easily accessed, managed and updated. Databases are essential because they allow businesses and individuals to store large amounts of data securely and retrieve it quickly.

Examples: - Banking systems store customer accounts and transactions
- Social media platforms store user profiles and messages

- ② A data warehouse is a centralised system designed for storing and analyzing large volumes of data from multiple sources. A data warehouse is optimised for analytics and reporting.

- ③ A data lake stores raw unstructured and structured data in its original format. Unlike a data warehouse, it doesn't require predefined schemas. Businesses use data lakes for flexibility of storing diverse data types.

- ④ A data mart is a smaller subset of a data warehouse designed for a specific department. It helps teams to quickly access data without searching the entire warehouse.

- ⑤ structured - organised in tables e.g. customer records in SQL
Semi-structured - Has some organisation but not in a table format e.g. JSON files.
Unstructured - unformatted data e.g. emails, videos and images

⑥ Data modeling is the process of designing how data will be structured and related in a system.

- e.g. ① Conceptual model
② Logical model

⑦ Data mining is the process of discovering patterns and insights from large datasets.
example ① Retailers analysing purchase history to recommend products.
② Banks detecting fraudulent transactions

⑧ Database management System

- MySQL: Open-source, good for web applications
- PostgreSQL: Advanced features, used for complex analytics
- Oracle: Enterprise level, handles big workloads
- MongoDB: NoSQL, great for unstructured data like documents

⑨ ETL

- Extract: pull data from sources
 - Transform: Clean and format data
 - Load: store data in a warehouse
- ETL ensures data is accurate and usable for analytics.

⑩ Data storage formats

- CSV: simple text, good for spreadsheets
- JSON: flexible, used in APIs.
- Parquet: Columnar, efficient for big data
- Avro: Compact, used in streaming systems.

SECTION B

⑪ AI is the simulation of human intelligence in machines

- Narrow AI: Specialised AI like Siri
- General AI: Human-like reasoning

⑫ ML is teaching computers to learn patterns from data instead of explicit programming

- Supervised: labeled data
- Unsupervised: finds patterns
- Reinforcement: learns by trial and error

⑬ Deep learning is a subset of ML using neural networks with many layers. Neural networks mimic how the brain processes information. They learn features automatically.

⑭ NLP enables machines to understand human language.

e.g. Spell check in word

- Google translate

- Chatbots like customer services assistants.

⑮ LLMs are advanced ML models trained on massive text datasets. They differ from standard ML by their scale and ability to generate human-like text.

e.g. GPT-4

⑯ Generative AI creates new content (texts, images and music)

e.g. 1. ChatGPT

2. DALL·E

3. GitHub

⑰ A training dataset is the data used to teach an AI model. Quality matters because biased or incorrect data leads to flawed predictions.

⑱ Computer Vision enables machines to interpret images

e.g. Self-driving cars detecting pedestrians

• Medical imaging detecting tumors

SECTION C

- 19) An API is a set of rules that lets software communicate e.g. A restaurant menu (API) tells you what dishes you can order.
APIs are vital for connecting apps and services.
- 20) API key is ~~used~~ a secret code that identifies and secures API usage. If exposed attackers could misuse the service.
- 21) REST API
REST = Representational state transfer methods:
- GET: Retrieve data
- POST: Add new data
- PUT: Update data
- DELETE: Remove data
- 22) JSON is a lightweight format of data exchange.
e.g. {"name": "Alice", "age": 25}
- 23) Prompt Engineering is crafting effective inputs for AI tools.
poor prompt: "Write something"
better prompt: "Write a 200-word summary of climate change impacts on farming"

(24) MAP applies a function to each item in a list.
e.g. numbers = [1, 2, 3]
squares = list(map(lambda x: x**2, numbers))

(25) Cloud Computing models:
- IaaS: Infrastructure (AWS EC2)
- PaaS: Platform (Google App Engine)
- SaaS: Software (Microsoft 365)

(26) Version control
GIT tracks code changes
- Commit: Save a snapshot
- Branch: Work on separate features

(27) Jupyter Notebook - An interactive environment mainly used in python. Its popular for data analysis because it combines code, visuals and notes.

(28) Python is popular for data analytics due to simplicity and libraries.
e.g. - Pandas (data manipulation)
- Numpy (numerical computing)
- Matplotlib (visualisation)
- Scikit-learn (machine learning)

SECTION 1

(29) Data privacy protects personal information.
GDPR is a European regulation that applies to EU countries and governs how data is collected, stored and shared.

(30) Data Ethics

Examples of unethical use:

- Selling user data without consent
- AI hiring tools discriminating against women
- Deepfake videos spreading misinformation

(31) Data governance is the framework for managing data responsibly. Key components:
- policies, roles, data quality, compliance.

(32) Bias occurs when AI produces unfair results.
eg. facial recognition misidentifying people.
consequences: wrongful arrests.

(33) Role	Focus	Skills	Tools
Data Analyst	Reports & dashboards	SQL, Excel	PowerBI, Tableau
Data Scientist	Predictive models	Python, ML	Jupyter, Scikit-learn
Data Engineer	Data pipelines	ETL, cloud Platforms	Spark, Hadoop
AI Engineer	AI Systems	ML, deep learning	Tensorflow, PyTorch

- (34) BI helps businesses make data-driven decisions
Tools : PowerBI
: Tableau
: Looker studio

SECTION 2

- (35) I found generative AI most interesting because it can create text, images, and music opening up endless creative possibilities
- (36) Deep learning was hardest to understand. I used Google's ML crash course and youtube tutorials to grasp neural networks
- (37) AI will affect my career by automating repetitive tasks and enabling smarter decision making.

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